

WISE: Self-Sustaining Population Equilibria over Decision-Induced Robot Networks

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Abstract—Distributed population-game dynamics need a connected communication graph to estimate the aggregates and pay-offs that define their equilibria. We study the self-referential case in which the equilibrium’s own spatial composition determines whether that graph exists. For a heterogeneous multi-robot team we show that the productively optimal aggregate is unique but its *composition* is degenerate—a positive-dimensional face of equal-payoff profiles—so classical theory fixes how much wrench each load receives, not which robot types provide it. On that face we place an endogenous, position-induced Laplacian and define the *Wrench- and Information-Self-Sustaining Equilibrium* (WISE): a productively optimal composition that also keeps algebraic connectivity above the Nash-seeking threshold $\sigma^* = \vartheta^2/(cm_F)$. We prove (i) composition degeneracy, (ii) that a self-sustaining equilibrium exists iff $\Lambda^* := \max_{x \in \mathcal{E}} \lambda_2(L(x)) \geq \sigma^*$, (iii) an exact local stability threshold $cm_F \lambda_2 > \vartheta^2$ derived from the coupled estimation dynamics, and (iv) that an infinitesimal information reward selects the most-connected composition without moving the productive optimum. Closed-loop simulation with rigid-load transport and second-order unicycles confirms all four, and shows relay roles emerging from radius heterogeneity by comparative advantage. A *population equilibrium can produce a configuration that destroys the graph its own algorithm requires; composition heterogeneity is exactly what lets it select the equilibria that survive.*

Index Terms—Multi-robot systems, aggregative games, generalized variational equilibrium, wrench feasibility, algebraic connectivity, coalition formation.

I. INTRODUCTION

Distributed population-game and Nash-seeking methods have become a standard route to coordinate large teams: agents adjust their strategies from locally communicated aggregates, and the dynamics converge to an equilibrium of a constrained potential game [1]–[4]. Every one of these methods rests on a silent premise: that the communication graph needed to *compute* the equilibrium is available while the equilibrium is being reached.

For a mobile robot team this premise is not innocent, because the agents that compute the equilibrium are the same agents whose placement forms the network. Consider a heterogeneous team recruited to move a load across a gap. A profile can be productively optimal—delivering exactly the demanded wrench (force and torque [5])—and yet commit every robot to lifting, so that no robot remains to relay, algebraic connectivity collapses, and the very estimator that produced the profile loses its consensus channel. The equilibrium destroys the means of its own computation.

The self-referential gap: The population-games literature treats the interaction graph as *given*: a constraint or resource external to the game [2], [3]. Connectivity-control and multi-robot task-allocation work maintains algebraic connectivity λ_2 as a separate design objective [6]–[10]. Neither addresses the case where the graph is an *output* of the equilibrium’s spatial composition. We adopt the distributed population-game and generalized-Nash machinery of Sandholm and of Barreiro-Gómez, Quijano, and Martínez-Piazuelo [1]–[4], and treat the self-referential case in which the decision itself generates the graph that machinery presupposes.

Key observation: The productive optimum pins down the aggregate but not the composition. Because the team payoff couples only through population aggregates [11], the set of profiles achieving the optimal aggregate is a positive-dimensional polytope (Theorem 1): classical theory says how much service each task gets, not *which types* provide it. That undetermined composition is exactly the freedom we use to keep the network alive.

Contributions:

- **Composition degeneracy** (Theorem 1): the productively optimal aggregate is unique, but the equal-payoff composition face $\mathcal{E}$ has dimension $(T-1)(K-1)$.
- **Self-sustaining existence** (Theorem 2): with an endogenous, position-induced Laplacian on $\mathcal{E}$, a self-sustaining equilibrium exists iff $\Lambda^* = \max_{x \in \mathcal{E}} \lambda_2(L(x)) \geq \sigma^*$. This is *selection within a degenerate Nash face*, not an added constraint.
- **Exact self-defeat threshold** (Theorem 3): the coupled strategy/consensus error is locally exponentially stable iff $cm_F \lambda_2 > \vartheta^2$, deriving $\sigma^* = \vartheta^2/(cm_F)$ rather than importing it.
- **Infinitesimal-reward selection** (Prop. 2) and a distributed algorithm, validated by closed-loop simulation with rigid-load transport and second-order unicycles—including emergent relay roles.

We name the resulting refinement the WISE equilibrium. A *WISE equilibrium can both execute itself and compute itself.*

II. POPULATION MODEL AND INDUCED NETWORK

A. A constrained potential population game

A team of N robots is partitioned into T types τ (heterogeneous in force capacity and communication range) with

masses m_τ . Each robot chooses an action from $\mathcal{A} = \{(k, h)\} \cup \{\text{relay}, \text{idle}\}$: a (load, slot) pair, relay, or idle. Following the population-game formalism [1], [2], the state is the mass distribution $x = (x_{\tau a})$ with $\sum_a x_{\tau a} = m_\tau$, i.e. $x \in \mathcal{X} = \prod_\tau m_\tau \Delta_{|\mathcal{A}|}$. We work with this *fluid relaxation*; $x_{\tau a}$ is a population mass, not a single robot's mixed strategy. Integer assignments are recovered by rounding, which preserves the certificates below whenever the feasibility margins exceed a rounding constant (Remark 1).

B. Directional support-capability map

Let $U_\tau \subset \mathbb{R}^2$ be the compact convex admissible-force set of a type- τ robot, $A_{kh} \in \mathbb{R}^{3 \times 2}$ the contact-to-wrench map at slot h of load k , and $\eta_{k\ell} \in \mathbb{R}^3$ a probing wrench direction. Define the *directional support-capability map*

$$\mathcal{W}_{\tau, kh\ell} = h_{U_\tau}(A_{kh}^\top \eta_{k\ell}), \quad (1)$$

with h_{U_τ} the support function of U_τ ; each entry is the largest force projection a type- τ robot can supply along the pulled-back direction. (We call (1) a *map*, not a tensor: it is a table of support values, not a multilinear object.) The population directional capacity and the projected demand are, writing $x_{\tau kh}$ for the mass of type τ on slot (k, h) ,

$$s_{k\ell}(x) = \sum_{\tau, h} \mathcal{W}_{\tau, kh\ell} x_{\tau kh}, \quad d_{k\ell} = \eta_{k\ell}^\top w_k^{\text{dem}}. \quad (2)$$

Proposition 1 (Support certificate for the relaxation). *Let $\mathcal{W}(C_k) = \{\sum_{\tau, h} A_{kh} f_{\tau h} : f_{\tau h} \in x_{\tau kh} U_\tau\}$ be the reachable population wrench set. For every finite direction set $\{\eta_{k\ell}\}$, $s_{k\ell}(x) \geq d_{k\ell} \forall \ell$ is necessary for $w_k^{\text{dem}} \in \mathcal{W}(C_k)$, and sufficient when $\{\eta_{k\ell}\}$ are the outer normals of $\mathcal{W}(C_k)$. If every U_τ is a polytope, $\mathcal{W}(C_k)$ is a zonotope and finitely many normals suffice; for smooth U_τ (e.g. disks) the finite test is an outer relaxation whose gap vanishes as the direction set refines.*

Proof. $\mathcal{W}(C_k)$ is compact convex, so by the supporting-hyperplane theorem $w_k^{\text{dem}} \in \mathcal{W}(C_k)$ iff $\eta^\top w_k^{\text{dem}} \leq h_{\mathcal{W}(C_k)}(\eta)$ for all η . Support functions are additive over Minkowski sums and positively homogeneous, giving $h_{\mathcal{W}(C_k)}(\eta_{k\ell}) = s_{k\ell}(x)$. A finite normal set yields the stated necessity/sufficiency; a polytopic U_τ makes $\mathcal{W}(C_k)$ a Minkowski sum of segments, i.e. a zonotope with finitely many facets. $\square$

We call load k *wrench-feasible* at x when $s_{k\ell}(x) \geq d_{k\ell}$ for all ℓ . Unlike a scalar capacity c_τ , (1) separates force limit, lever arm, and slot—the channels that distinguish otherwise identical types.

C. Endogenous, decision-induced Laplacian

Only robots that do not lift can relay. Let $\mathcal{C}_{\tau re} \geq 0$ be the contribution of a type- τ mass in role r to communication link e ; edge weights and the induced Laplacian are affine in the decision,

$$a_e(x) = \sum_{\tau, r} \mathcal{C}_{\tau re} x_{\tau r}, \quad L(x) = \sum_e a_e(x) L_e. \quad (3)$$

The coefficients $\mathcal{C}_{\tau re}$ encode which candidate sites a relaying type can bridge, so (3) is the convex, decision-induced relaxation of the position-dependent graph $L(q)$; the continuous-position geometry $L(q(x))$ is nonconvex and is exercised only in the closed-loop simulation (Section V). Let $Q \in \mathbb{R}^{V \times (V-1)}$ be an orthonormal basis of $\mathbf{1}^\perp$; the information constraint is the LMI

$$Q^\top L(x) Q \succeq \sigma^* I \iff \lambda_2(L(x)) \geq \sigma^*, \quad (4)$$

affine in x , with PSD matrix dual $Z \succeq 0$; at a simple minimiser $Z = \pi v v^\top$ recovers a scalar connectivity price π (Section IV). The threshold σ^* is *derived*, not assumed, in Theorem 3.

Lemma 1 (Convex information set). $x \mapsto \lambda_2(L(x))$ is concave, so (4) defines a convex set.

Proof. $\lambda_2(L) = \min_{\|v\|=1, v \perp \mathbf{1}} v^\top L v$ is a pointwise minimum of functions linear in L [6], hence concave; composed with the affine $L(x)$ it stays concave, and its super-level set is convex [7]. $\square$

III. SELF-SUSTAINING EQUILIBRIA

A. Productive value and the composition face

The team's productive value couples through aggregates only,

$$V(x) = \sum_k F_k(s_k(x)) - \sum_{\tau, a} g_{\tau a} x_{\tau a}, \quad (5)$$

with F_k strictly concave and non-decreasing (a canonical choice is the deficit value $F_k(s_k) = -\frac{1}{2} \|d_k - s_k\|_{Q_k}^2$, $Q_k \succ 0$) and $g_{\tau a} \geq 0$ the action cost. This is a *constrained potential population game*: the type-action payoff is $\partial V / \partial x_{\tau a}$, so V is an exact potential by construction [1], [12]. We do not claim novelty here—the game is identical-interest—the novelty is what V leaves undetermined.

Let $y = Bx$ collect the aggregates $s(x)$ appearing in (5), with B linear, and let $X_f = \{x \in \mathcal{X} : s_{k\ell}(x) \geq d_{k\ell}\}$ be the wrench-feasible set. Write $V(x) = \Phi_0(Bx) - \langle g, x \rangle$.

Theorem 1 (Composition degeneracy). *Let Φ_0 be strictly concave and g lie in the row space of B (cost expressible through aggregates). Then the productively optimal aggregate $y^* = Bx^*$ is unique, but the optimal set*

$$\mathcal{E} = \{x \in X_f : Bx = y^*\} \quad (6)$$

is a polytope of dimension $(T-1)(K_{\text{act}}-1)$, where K_{act} is the number of active (load, slot) columns.

Proof. Strict concavity of Φ_0 makes the maximiser y^* of $\max_y \{\Phi_0(y) - (\text{aggregate cost}) : y \in BX_f\}$ unique. $\mathcal{E}$ is the preimage $B^{-1}(y^*) \cap X_f$, an affine slice of the product of type simplices with fixed row sums m_τ and fixed column aggregates y^* —a transportation polytope, whose dimension is $(T-1)(K_{\text{act}}-1)$ [1]. Since g acts through B , every $x \in \mathcal{E}$ attains the same V . $\square$

Classical theory thus fixes *how much* wrench each load receives, not *which types* provide it. The face $\mathcal{E}$ is the freedom we exploit.

B. The WISE refinement

Place the endogenous Laplacian (3) on $\mathcal{E}$.

Definition 1 (WISE). A profile $x^* \in \mathcal{E}$ is a Wrench- and Information-Self-Sustaining Equilibrium if $\lambda_2(L(x^*)) \geq \sigma^*$, i.e. $x^* \in \mathcal{E}_{ss} := \mathcal{E} \cap \{x : \lambda_2(L(x)) \geq \sigma^*\}$.

Theorem 2 (Existence by selection). Let $\Lambda^* = \max_{x \in \mathcal{E}} \lambda_2(L(x))$. Then

$$\mathcal{E}_{ss} \neq \emptyset \iff \Lambda^* \geq \sigma^*, \quad (7)$$

and $\mathcal{E}_{ss}$ is convex. When $\dim \mathcal{E} > 0$ and $\min_{\mathcal{E}} \lambda_2 < \sigma^* \leq \Lambda^*$, the refinement is strict: some productively optimal compositions self-sustain and others do not.

Proof. $\lambda_2 \circ L$ is concave (Lemma 1) and continuous on the compact convex $\mathcal{E}$, so it attains its maximum Λ^* ; $\mathcal{E}_{ss}$ is nonempty iff $\Lambda^* \geq \sigma^*$. $\mathcal{E}_{ss}$ is the intersection of the polytope $\mathcal{E}$ with a convex super-level set, hence convex. Strictness is immediate when the threshold falls strictly inside the range of λ_2 over $\mathcal{E}$. $\square$

Selection within a degenerate Nash face is stronger than appending a constraint to a strictly concave program: without Theorem 1 there would be a single composition and nothing to select.

C. The self-defeat threshold, derived

The distributed dynamics run a strategy-revision loop and a consensus estimator of the aggregates over $L(x)$. Linearising about x^* , with m_F the strong-monotonicity constant of the revision protocol, c the consensus gain, and ϑ the cross-coupling by which stale aggregate estimates perturb revision, the coupled error $e = (e_x, e_\eta)$ obeys

$$\dot{e} = J e, \quad J = \begin{bmatrix} -m_F & \vartheta \\ \vartheta & -c \lambda_2(L(x^*)) \end{bmatrix}. \quad (8)$$

Theorem 3 (Exact stability threshold). x^* is locally exponentially stable under (8) iff

$$c m_F \lambda_2(L(x^*)) > \vartheta^2, \quad \text{i.e.} \quad \lambda_2 > \sigma^* := \frac{\vartheta^2}{c m_F}. \quad (9)$$

Below the threshold J has a positive real eigenvalue and x^* is unstable.

Proof. $\text{tr } J = -(m_F + c \lambda_2) < 0$ always, and $\det J = c m_F \lambda_2 - \vartheta^2$. A real 2×2 matrix is Hurwitz iff $\text{tr } J < 0$ and $\det J > 0$; thus stability $\iff \det J > 0 \iff$ (9). If $\det J < 0$ the eigenvalues are real of opposite sign, giving an unstable manifold. $\square$

This grounds σ^* in the dynamics of the very estimator that computes the equilibrium, and matches the existence condition (7): a productively optimal composition is dynamically viable iff its induced connectivity clears the same σ^* .

D. Selection, prices, and the cost of endogeneity

Proposition 2 (Infinitesimal-reward selection). For $V_\varepsilon(x) = V(x) + \varepsilon \lambda_2(L(x))$, $\varepsilon > 0$, every maximiser x_ε over X_f satisfies $Bx_\varepsilon \rightarrow y^*$ and, along a subsequence, $x_\varepsilon \rightarrow \arg \max_{x \in \mathcal{E}} \lambda_2(L(x))$ as $\varepsilon \downarrow 0$.

Proof. V is upper bounded on the compact X_f with optimal set $\mathcal{E}$ (Theorem 1); $\lambda_2 \circ L$ is continuous. By the Tikhonov/ Γ -selection principle, maximisers of $V + \varepsilon \lambda_2$ converge as $\varepsilon \downarrow 0$ to maximisers of λ_2 over $\arg \max V = \mathcal{E}$. $\square$

An infinitesimal information value breaks the composition degeneracy toward the most-connected profile without moving the productive optimum—an exact equilibrium refinement.

Corollary 1 (Two certification prices). At a WISE solving $\max_{x \in \mathcal{E}_{ss}} V$, let $\mu_k^* = \nabla F_k(s_k^*)$ be the wrench price and $Z^* \succeq 0$ the dual of the LMI (4), with $Z^* = \pi^* v v^\top$ at a simple minimiser. Stationarity gives the type-action value

$$q_{\tau a}^* = \underbrace{\sum_{k,\ell} \mu_{k\ell}^* \mathcal{W}_{\tau a, k\ell}}_{\text{wrench}} - g_{\tau a} + \underbrace{\pi^* \partial \lambda_2 / \partial x_{\tau a}}_{\text{connectivity}}, \quad (10)$$

with $\pi^*(\lambda_2 - \sigma^*) = 0$. All terms of ∇V are present: $\mu^* = \nabla F_k$ is the single wrench price, not a separate multiplier.

We measure the price of endogeneity *additively*, avoiding the pathologies of a potential ratio:

$$\Delta_{\text{endo}} = V(x_{\text{exo}}^*) - \max_{x \in \mathcal{E}_{ss}} V(x) \geq 0, \quad (11)$$

the productive value forgone to remain self-sustaining. $\Delta_{\text{endo}} = 0$ exactly when Λ^* is attained at a productively optimal composition—when radius heterogeneity suffices to bridge the gap for free.

Remark 1 (Integer recovery). Rounding the fluid optimum x^* to integer assignments preserves wrench feasibility and $\lambda_2 \geq \sigma^*$ whenever the margins $m_W = \min_{k,\ell} (s_{k\ell} - d_{k\ell})$ and $m_I = \lambda_2 - \sigma^*$ exceed rounding constants $\kappa_W, \kappa_I = O(1/N)$ set by the granularity.

IV. DISTRIBUTED ALGORITHM

We seek a WISE by a projected primal-dual flow that couples a population revision protocol to the connectivity price of Corollary 1:

$$\dot{x}_\tau = \mathcal{R}_\tau \left(x, \nabla_{x_\tau} V(x) + \pi \nabla_{x_\tau} \lambda_2(L(x)) \right), \quad (12a)$$

$$\dot{\pi} = [\sigma^* - \lambda_2(L(x))] \pi, \quad (12b)$$

where $\mathcal{R}_\tau$ is a mass-preserving revision protocol and the wrench price $\mu_k = \nabla F_k(s_k)$ is internal to ∇V . Each type needs only its own capability rows $\mathcal{W}_{\tau,\cdot}$, the aggregate deficit $d - s(x)$, and the local Fiedler sensitivity $v^\top L_e v$, all obtained by consensus over the current graph. Economically: a missing wrench direction raises μ ; missing connectivity raises the relay price π ; mass migrates toward the highest type-action value (10).

Choice of revision protocol: We use a projected-ascent (Euclidean) protocol, which is Nash-stationary: its rest points are exactly the constrained maximisers of V [1]. Smith and BNN protocols share this property. We deliberately avoid the replicator (multiplicative-weights) map: it is *not* Nash-stationary on the boundary—it can rest at support-restricted profiles that leave a profitable but extinct action unused, and it never repopulates a mass it has driven to zero [1]. On a face $\mathcal{E}$ whose self-sustaining subset may require re-activating a relay type, replicator can therefore stall short of $\mathcal{E}_{ss}$; projected ascent cannot.

Proposition 3 (Convergence to a WISE). *Assume Φ_0 strongly concave, Slater feasibility of $\mathcal{E}_{ss}$, and a Nash-stationary revision $\mathcal{R}$. Then the saddle points of the Lagrangian of $\max_{x \in \mathcal{E}_{ss}} V$ coincide with the WISE profiles, and (12) is a projected monotone flow whose trajectories converge to one, provided the consensus gain keeps the estimator above the stability threshold of Theorem 3 throughout.*

Sketch. Strong concavity of V and convexity of $\mathcal{E}_{ss}$ make the Lagrangian concave-convex, so its saddle points are the KKT (hence WISE) points by Corollary 1. The map in (12) is the projected gradient of this saddle function, which is monotone; passivity arguments for primal-dual Nash-seeking give convergence [13], [14]. The proviso is exactly (9): if the trajectory drops below σ^* the estimator error grows (Theorem 3) and the aggregates fed to $\mathcal{R}$ are no longer trustworthy—the algorithm can only be guaranteed to converge where the equilibrium is self-computing. $\square$

The proviso is the paper’s point in algorithmic form: convergence of the distributed method is contingent on the equilibrium it is converging to being information self-sustaining.

V. SIMULATION STUDY

We validate the four results on heterogeneous teams ($N = 12$ unless noted) in a two-region world: a rigid load in the right cluster demands a planar wrench, and short links connect robots only within a cluster, so the gap is bridged only when a long-range type relays. The strategic layer runs the primal-dual flow (12); the physical layer integrates a rigid load and second-order unicycles, with each lifter’s contact force confined to a heading-aligned elliptical set (nonholonomic: free along the heading, limited laterally), so wrench realizability genuinely depends on pose. The primary metric is the *certified service rate*: a load is served only if it is wrench-feasible and $\lambda_2 - \sigma^* \geq 0$. Every instance is seeded and reproduces exactly.

E1 — Convergence (Fig. 2): On a representative instance the distributed flow drives $\lambda_2(L(x))$ onto σ^* and the wrench residual to zero together: the equilibrium becomes wrench-feasible and self-computing at once. The information price π settles at its certification value and the relay mass accrues *only* to long-range types.

E2 — Self-defeat threshold (Fig. 3, Thm. 3): On one instance we compare two compositions from the same face $\mathcal{E}$. When a long-range robot relays, the induced $\lambda_2(t)$ rises

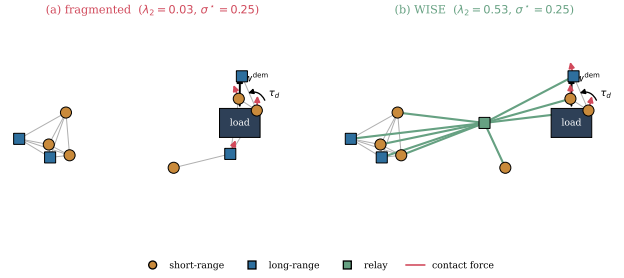


Fig. 1. The paradox, physically. (a) A productively optimal composition commits long-range types to their clusters; short links leave the gap open, $\lambda_2 < \sigma^*$, and the estimator that produced the composition loses consensus. (b) A WISE composition on the same face routes one long-range robot to relay, restoring $\lambda_2 \geq \sigma^*$ while the lifters still realise the demanded wrench (red contact forces).

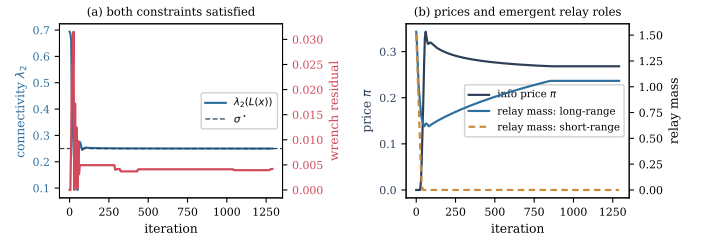


Fig. 2. Distributed convergence. (a) $\lambda_2 \rightarrow \sigma^*$ and wrench residual $\rightarrow 0$ together. (b) Price π converges; relay mass accrues to long-range types only (short-range stays at zero).

above σ^* and the coupled estimation error $\|e(t)\|$ decays; when it is withdrawn, $\lambda_2(t)$ stays below and $\|e(t)\|$ grows—the equilibrium defeats its own estimator. The sign change occurs exactly at $c m_F \lambda_2 = \vartheta^2$, the derived threshold, not a fitted curve.

E3 — Existence phase diagram (Fig. 4, Thm. 2): Sweeping the long-range fraction ν and torque demand τ_d (12 seeds/cell), certification collapses exactly where predicted: at $\nu \rightarrow 0$ no composition on $\mathcal{E}$ reaches $\Lambda^* \geq \sigma^*$ (existence fails), and at high τ_d the fleet is wrench-limited. The certified band is the region where $\Lambda^* \geq \sigma^*$.

E4 — Constraint ablation (Table 1, Fig. 5a): On paired seeds, enforcing either requirement alone fails: `scalar_capacity` and `wrench_only` build valid coalitions that fragment the network, `connectivity_only` keeps the network but cannot lift, and only WISE certifies both—matching the centralized oracle, evidence that the distributed selection attains $\max_{x \in \mathcal{E}_{ss}} V$.

E5 — Emergent roles (Fig. 5b, Prop. 2): Pooling WISE solutions across seeds, relay commitment is drawn *exclusively* from long-range types: no short-range robot ever relays. The price system (10) selects, within the degenerate face $\mathcal{E}$, the composition that maximises connectivity—comparative advantage in radius produces the division of labour, exactly the infinitesimal-reward selection of Proposition 2.

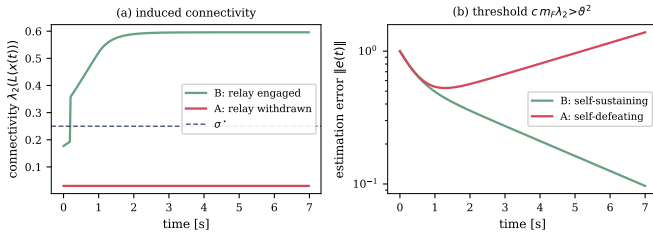


Fig. 3. Closed-loop self-defeat. (a) Induced connectivity for a self-sustaining (B) vs. self-defeating (A) composition. (b) Estimation error decays iff $c m_F \lambda_2 > \psi^2$ (Theorem 3); it diverges below the threshold.

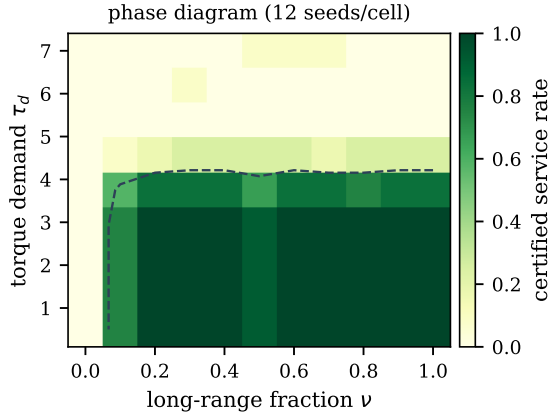


Fig. 4. Certified-service-rate phase diagram over (ν, τ_d) (12 seeds/cell; dashed = 50% contour). Certification requires $\Lambda^* \geq \sigma^*$ (fails at $\nu \rightarrow 0$) and wrench feasibility (fails at high τ_d).

VI. CONCLUSION

We studied population equilibria whose own spatial composition determines whether the communication graph their algorithm requires exists. Three facts make the phenomenon tractable and sharp: the productive optimum fixes the aggregate but leaves a positive-dimensional composition face (Theorem 1); on that face a self-sustaining equilibrium exists iff its best induced connectivity clears the Nash-seeking threshold (Theorem 2); and that same threshold is the exact stability boundary of the coupled estimator (Theorem 3), so an infinitesimal information reward selects the surviving composition (Proposition 2). Closed-loop simulation with rigid-load transport and second-order unicycles confirms all four and shows relay roles emerging by comparative advantage. We extend the distributed population-game and generalized-Nash machinery of the Sandholm and Quijano–Barreiro–Gómez–Martínez-Piazzuelo lines [1]–[4] to the case where the decision generates the graph that machinery presupposes.

Limitations: The analysis is a *fluid relaxation*; integer assignments are recovered by rounding under the margin condition of Remark 1. The induced Laplacian is affine in the decision (candidate-site relaxation of the position-dependent graph); the fully continuous-position geometry $L(q)$ is non-convex and is exercised only in simulation. Nonholonomic wrench realizability is modelled by the heading-aligned force

TABLE I
CONSTRAINT ABLATION ON PAIRED SEEDS ($N = 12$, $\nu = 0.5$, $\tau_d = 3$, SEED-AVERAGED). ARROWS MARK THE PREFERRED DIRECTION.

Method	Wr. res. $\downarrow$	$\lambda_2 - \sigma^* \uparrow$	Cert. succ. $\uparrow$
scalar_capacity	0.00	-0.22	0.0%
wrench_only	0.02	-0.18	6.2%
connectivity_only	2.11	+0.02	0.0%
wise_pd	0.02	+0.02	93.8%
centralized (oracle)	0.02	+0.02	93.8%

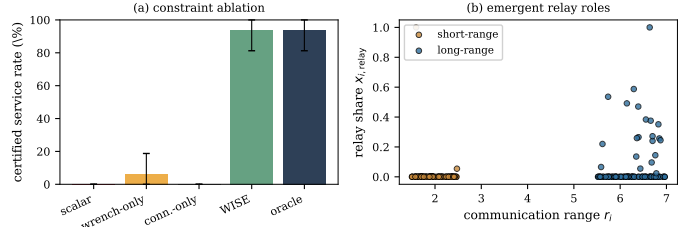


Fig. 5. (a) Certified service rate by method (mean $\pm$ 95% bootstrap CI): single-requirement methods fail; WISE matches the oracle. (b) Relay share vs. communication range at the WISE solution, pooled over seeds—relays are drawn only from long-range types.

set, not proved; and validation is in simulation.

Future work: A realizability theorem for the nonholonomic contact, the nonconvex continuous-position graph, and hardware validation are the natural next steps toward a full closed-loop guarantee.

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